MIRMIS/NIR–MIR Calibration pipeline description

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Applicable Documents

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AD-01	MIRMIS Instrument Design Report
AD-02	Hera Image Format and Calibration Levels
AD-03	ASP Calibration Report v0.81

Reference Documents

[RD1] Andrew M. Annex et al. "SpiceyPy: a Pythonic Wrapper for the SPICE Toolkit". In: *Journal of Open Source Software* 5.46 (2020), p. 2050. DOI: 10.21105/joss.02050. URL: https://doi.org/10.21105/joss.02050.

[RD2] Lucio Chiappetti et al. *Definition of the Flexible Image Transport System (FITS): The FITS Standard Version 4.0*. Tech. rep. Accessed: 2024-12-20. NASA/IAU, 2018. URL: https://fits.gsfc.nasa.gov/fits_standard.html.

[RD3] SemiConductor Devices. *SWIR640 FPA User Guide*. 2020.

[RD4] Geraint H Jones et al. "Comet Interceptor Mission". In: *Encyclopedia of Astrobiology* (2023), pp. 641–642.

[RD5] Tomas Kohout et al. "Feasibility of asteroid exploration using CubeSats—ASPECT case study". In: *Advances in Space Research* 62.8 (2018), pp. 2239–2244.

[RD6] P. Michel et al. "The Hera Space Mission in the Context of Small Near-Earth Asteroid Missions in the Past, Present and Future". In: *Space Science Reviews* 221.70 (2025). DOI: 10.1007/s11214-025-01195-1.

[RD7] NASA Jet Propulsion Laboratory. *Navigation and Ancillary Information Facility (NAIF)*. https://naif.jpl.nasa.gov/naif/index.html. Accessed: March 03, 2025 (no publication date). n.d.

[RD8] A. Näsilä et al. "The Asteroid Spectral Imager (ASPECT) on the Milani CubeSat". In: *Space Science Reviews* (2025). accepted.

[RD9] Antti Näsilä and Tomas Kohout. "Miniaturized spectral imaging instrumentation for planetary exploration". In: *2020 IEEE Aerospace Conference*. IEEE. 2020, pp. 1–7.

[RD10] NOAA National Centers for Environmental Information. *Solar Spectral Irradiance Climate Data Record, Version 3.0*. https://www.ncei.noaa.gov/products/climate-data-records/solar-spectral-irradiance. accessed 2025/02/21. 2025. DOI: doi:10.25921/esjz-1w61.

Acronyms

ASPECT	Asteroid Spectral Imager
AIS	Asteroid Imaging Simulator
ESA	European Space Agency
FITS	Flexible Image Transport System
FWHM	Full Width Half Maximum
HDU	Header and Data Unit
MIRMIS	Modular Infrared Molecules and Ices Sensor
MIR	Mid infrared single-point spectrometer channel on MIRMIS
NIR	Near infrared hyperspectral imager channel on MIRMIS
NASA	National Aeronautics and Space Administration
NOAA	National Centers for Environmental Information
PDS4	Planetary Data System version 4
TIRI	Thermal infrared imager channel on MIMRIS
VTT	VTT Finland

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1. Introduction

This document describes the preliminary design and implementation of the data processing pipeline for the MIRMIS instrument NIR and MIR channels. MIRMIS instrument package is being developed for the ESA Comet Interceptor mission. As part of the wider Comet Interceptor payload, it is developed to meet the mission's top-level science goal of understanding the diversity of comets by mapping the composition and temperature of the nucleus and coma [AD-01].

The MIRMIS instrument covers spectral range 0.9–25 μm using three channels: NIR, hyperspectral imager (0.9–1.7 μm); MIR, single-point spectrometer (2.5–5.0 μm); and TIRI, multispectral thermal imager (6–25 μm) [RD4]. The MIR and NIR modules are based on tunable Fabry-Pérot interferometers which are used as an adjustable bandpass filter [RD9]. The NIR design is based on refractive optics with heritage from the ASPECT imager in the Milani CubeSat for ESA's Hera mission [RD5, RD6]. MIRMIS/NIR and MIRMIS/MIR are supplied by VTT Finland.

The MIRMIS/NIR–MIR data processing pipeline was developed to facilitate an accurate ground calibration of the acquired instrument data and generate ESA/NASA standard compliant data products (FITS/PDS4).

This document outlines the process and methodologies within the pipeline, which transforms raw instrument data into a single FITS file, containing a calibrated hyperspectral image of the target, complete with metadata about the acquisition. The result FITS file can be easily converted into a PDS4 product.

2. Overview

The pipeline is organized into distinct processing levels, each corresponding to a specific stage of the calibration or analysis. Each processing level consist of one or more steps. Beginning with raw data, each level progressively applies data calibration or processing procedures. Each processing level outputs its results in a FITS file. Level 0 creates a FITS file from the raw binary data, while level 1 focuses on calibrating and correcting the data. Both levels are applied individually for NIR and MIR channels.

The Figure 1 outlines processing levels applicable to each channel. Columns represents the individual imaging channels (NIR and MIR) while the rows correspond to a specific processing steps associated with a particular processing levels. A solid square within the diagram indicates that the corresponding step of the respective level is applicable to the particular channel. Each processing level builds upon the output of the preceding level. After each level (0A, 1A, 1B, and 1C) a new FITS file is created and saved.

2.1. Program and libraries

The data processing pipeline is developed with the Python programming language. Python is widely used in scientific computing and data processing due to its large ecosystem of libraries. Python provides libraries for data manipulation and image processing, which are essential for the tasks in this pipeline. The core libraries used in the pipeline are: Numpy (numerics), Astropy (FITS I/O and headers), SpicePy (SPICE data). For the full list of dependencies, see pyproject.toml in the repository.

2.2. Running the pipeline

The data from each acquisition is expected to arrive as binary files and JSON metafiles. Each exposure produces a separate binary file, while the associated metadata are provided in JSON format. The expected input directory structure is illustrated in Figure 2. The input directory is organized into two main subdirectories. The meta directory contains the metadata required to build the FITS headers and must include at least config.json and telemetry.json. The acquisition

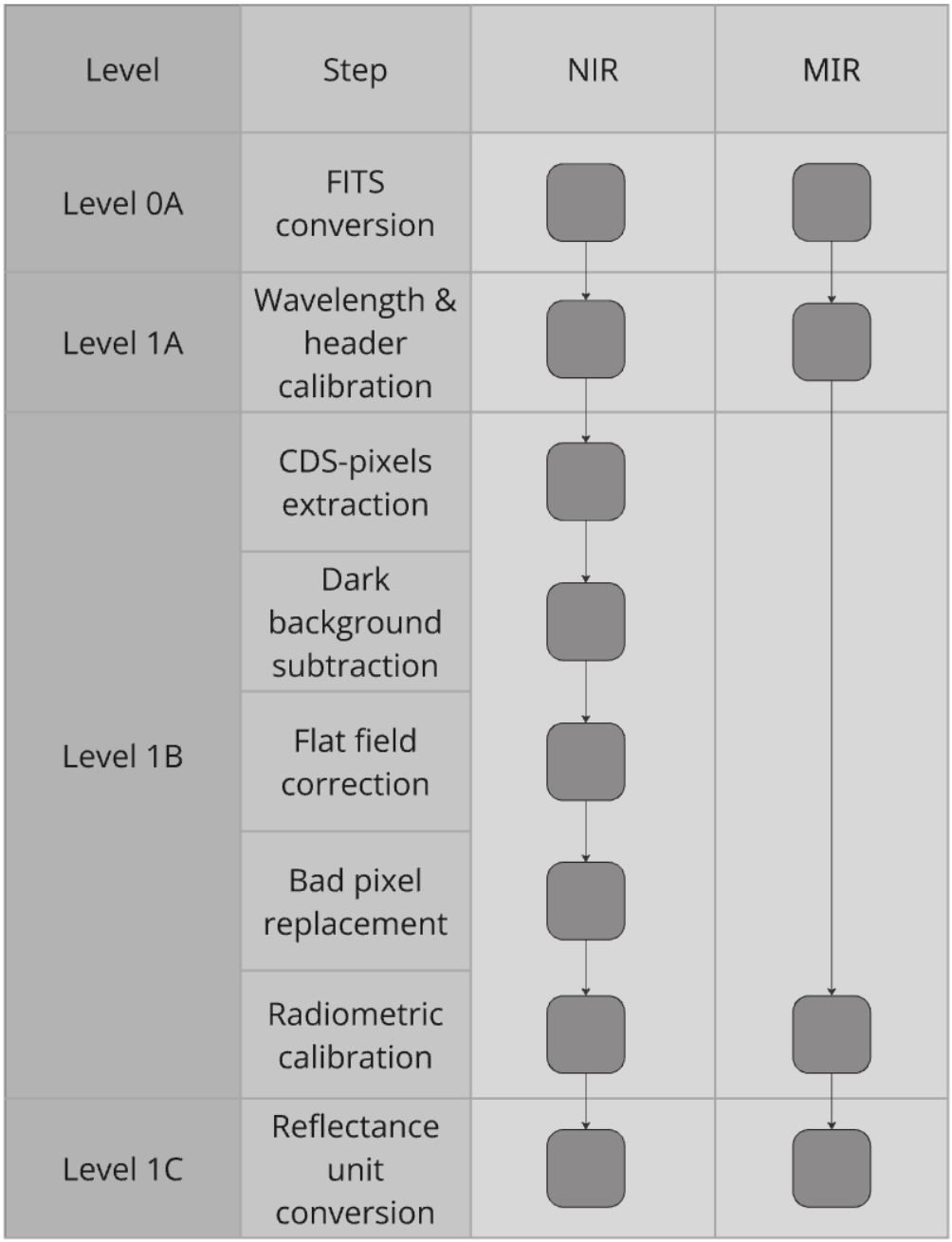

Figure 1: The pipeline with processing done in each data level. Columns represent the NIR and MIR channels and a square on particular data level indicates that corresponding procedure is applied to that channel.

data are stored in a directory whose name begins with 'acq_';

Image files are stored in binary format and the expected naming convention is based on ASPECT data files naming defined in [AD-02]. Character 4 specifies the channel index—0 for NIR, and 1 for MIR. Characters 10-12 indicates the exposure number of that acquisition. For example, For

example, dc_0_exp_000.bin corresponds to the first frame of NIR channel. Files may appear either in raw binary format or as JPEG2000-decoded images with the .jp2 suffix.

```
input_directory/
├── acq_000/
│   ├── dc_0_exp_000.bin.jp2
│   ├── dc_0_exp_001.bin.jp2
│   ├── dc_3_exp_000.bin
│   └── ...
├── meta/
│   ├── config.json
│   ├── telemtetry.json
│   └── ...
└── meta_diff_encoding (optional)/
    └── diff_encodings.json
```

Figure 2: Input directory structure expected by the pipeline

The FITS file created at each processing level are stored into an output directory. The naming convention of the pipeline results and the structure of each processing level follow the guidelines defined in [AD-02] so the imaging channel and processing level can be easily identified from the FITS file name. Characters 1-4 specify the channel data included. Characters 5-10 represent an unique image number, increasign by the acquisition time. It is generated as the base 32 coding of the image capture SC clock seconds. Characters 12-24 are teh date and time of the exposure UTC: e.g. 230131T140112_ for UTC 2023-01-31T14:01:12. Characters 26-27 of the filename indicate the processing level. For example, NIR_009FEQ_270323T060000_1A.fits refers to a data product stored after completing the level 1A. Detailed instructions to run the pipeline are provided in the README.md file of the repository.

2.3. Data products

All FITS files created by the pipeline conform to the FITS standard version 4.0, as defined in [RD2]. FITS file structure consists of the mandatory primary header and data unit (HDU), followed by optional extensions or special records. In this pipeline, the main data are stored in the Primary HDU. This consisting of all relevant metadata, and a data unit containing either a three-dimensional hyperspectral data cube for NIR or an one-dimensional array for MIR derived from the corresponding raw acquisition binary files. For NIR, an additional Binary Table extension

is included from processing level 1B onward. This extension stores the CDS pixels extracted from the original near-infrared images [RD3].

2.3.1 Primary HDU

The Primary HDU contains all metadata related to the image acquisition. This includes the information about the acquisition, relevant SPICE metadata, calibration-specific metadata, and the instrument channel specific metadata. In accordance with the FITS standard, the Primary header begins with a set of mandatory default entries, followed by the acquisition metadata entries.

2.3.2 Binary Table HDU

Following the Level 1B processing the FITS files containing NIR data will include an additional Binary Table extension. This extension stores the CDS pixels that are extracted from each NIR frame.

3. Pipeline levels

This section provides an overview of how each processing level transforms the NIR-MIR data. The format and data type of the data vary depending on the channel characteristics and the processing level applied as illustrated in the Figure 3. The amount of the image frames per instrument channel, denoted by n the table, may also vary.

3.1. Level 0

The level 0A converts raw instrument data into FITS format.

The level 0A combines the exposure frames of a single channel to a single data product stored in the Data Unit of the Primary HDU, with the metadata retrieved from the same input directory stored in the Primary Header.

Processing level	NIR	MIR
0A	(n x 518 x 648) uint16	(n,) uint32
1A	(n x 518 x 648) uint16	(n,) uint32
1B	(n x 512 x 640) float32 (>f4)	(n,) float32 (>f4)
1C	(n x 512 x 640) float32 (>f4)	(n,) float32 (>f4)

Figure 3: The Data Unit shape and data type by processing level.

3.1.1 SPICE Metadata Retrieval

The SPICE Kernel Dataset contains ancillary information about various objects, spacecrafts, and instruments of the mission. Most notably it provides geometric and positional information of the target which can be queried based on e.g. time, frame, and observer [RD7]. The spice metadata is appended to the Primary header at the processing level 0A.

The metadata that is retrieved from SPICE includes spacecraft position, image target position and distance, sun position, solar distance and elongation, earth position and distance, spacecraft quaternions (angle), and SPICE version. The queries are implemented in Python with the usage of SpiceyPy [RD1]. During the pipeline development of this pipeline, no meta kernel yet exist for Comet Interceptor mission, thus the part is inherited from the Hera mission ASPECT pipeline and likely some parts will need to be updated later.

3.1.2 Header

In accordance with the FITS standard, the Primary header begins with a set of mandatory default entries, followed by the acquisition metadata entries specified in the Table 1. Certain instrument metadata fields are specific to individual channels and exposures. These entries have `<channel>` and `<exp_id>` to indicate the corresponding channel and exposure number respectively.

The instrument-specific include the key temperature readings. The CCDTMP (detector temperature), FPI1 (spectral filter temperature 1), and FPI2 (spectral filter temperature 2) are provided for each channel, even if no other data from that channel is included in the data product. The channels present in the data product are listed as a comma-separated string in the 'CHANNELS' header entry.

Table 1: Primary header metadata entries.

Key	**Description**	Type
INSTRUME	Camera ID, 'MIRMIS'	String
ORIGIN	'ESA-COMET-INTERCEPTOR'	String
MISSPHAS	Mission Phase ID	String
OSERV_ID	Observation ID	String
FILENAME	Name of the FITS file	String
ORIGFILE	Original filename, e.g. 'dc_X_exp_XXX.bin'	String
SWCREATE	Software identification, e.g. 'NIR-MIRCAL v1.0'	String
DATE	UTC time of file creation, e.g. '2025-07-31T15:13:47'	String
PROCLEVL	Current calibration level	String
DATE-OB	UTC time of observations	String
SC_CLK	Spacecraft clock instrument format	String
OBJECT	Observation object	String
SPICE_MK	SPICE metakernel	String
SPICEVER	SPICE dataset version	String

Continued on next page

Table 1: Primary header metadata entries.

Key	**Description**	Type
SPICECLK	SC clock SPICE format, e.g. '1/077436069.1856'	String
SUN_POSX	Sun position vector X [km]	String
SUN_POSY	Sun position vector Y [km]	String
SUN_POSZ	Sun position vector Z [km]	String
SOLAR_D	Solar distance [AU]	String
EARTPOSX	Earth position vector X [km]	String
EARTPOSY	Earth position vector Y [km]	String
EARTPOSZ	Earth position vector Z [km]	String
EARTH_D	Earth distance [AU]	String
TARGET	Observation target (SPICE)	String
TRG_POSX	Target position vector X [km]	String
TRG_POSY	Target position vector Y [km]	String
TRG_POSZ	Target position vector Z [km]	String
TRG_DIST	Target distance [AU]	String
SC_QUATW	Spacecraft quaternion (W)	String
SC_QUATX	Spacecraft quaternion (X)	String
SC_QUATY	Spacecraft quaternion (Y)	String
SC_QUATZ	Spacecraft quaternion (Z)	String
CAM_RA	Camera axis RA [deg]	String
CAM_DEC	Camera axis DEC [deg]	String
CAM_NAZ	Camera axis north azimuth [deg]	String
SOL_ELNG	Solar elongation	String
CHANNELS	Channel data in this file, e.g. 'NIR'	String
NIR_CCDTMP	NIR Detector temperature	String
NIR_FPI1	NIR FPI 1 temperature	String

Continued on next page

Table 1: Primary header metadata entries.

Key	**Description**	Type
NIR_FPI2	NIR FPI 2 temperature	String
MIR_CCDTMP	MIR Detector temperature	String
MIR_FPI1	MIR FPI 1 temperature	String
MIR_FPI2	MIR FPI 2 temperature	String
`<channel>`_FRAMES	All frames included, comma-separated if multiple	String
`<channel>`_TASK_NUMBER	Number of tasks	String
`<channel>`_TASK_`<exp_id>`	'sp1 sp2 sp3 exposure'	String

3.2. Level 1

The purpose of level 1 is to calibrate and correct the data so that it accurately represents the information about the target in scientifically meaningful units. It ensures that the instrument's raw measurements are translated into values that can be reliably used in scientific analysis, reducing errors caused by instrument imperfections. The level is divided into 1A, 1B, and 1C. Level 1A converts metadata values like temperatures from DN values to kelvins and calculates the wavelengths from setpoint values. Level 1B applies multiple correction procedures to the data, namely, extracting diagnostic pixels, dark background subtraction, flat field correction, bad pixel replacement and radiometric calibration. Level 1C converts the radiance values into reflectance units.

3.2.1 Level 1A

The objective of header and wavelength calibration is to convert instrument temperature readings from raw DN values to calibrated temperatures in Kelvin, to convert the exposure DN values to seconds and to convert the piezo setpoint values into actual wavelength values (nm). Both channels utilizes a tunable Fabry-Pérot interferometer (FPI) developed at VTT Technical Research Center of Finland Ltd.. The FPI consist of two parallel reflective surfaces, which causes the light to reflect back and forth, creating a constructive and deconstructive interface. Essentially, the

FPI can be used as a bandpass filter, where a specific wavelength can be determined based on the tunable distance between the reflective surfaces. The distance is controlled in a closed loop, where the FPI control electronics measure the capacitance between three pairs of electrodes deposited on the mirror substrate [RD8]. The set point readings are stored into a task file from where the values can be converted into corresponding wavelengths by a mathematical formula 1, where sp is the set point value while A and B are constants determined during the calibration procedure. The value C is a correction value that can be modified based on wavelength calibration.

The converted wavelengths and exposures for each exposure are added to the header as individual entries. The naming convention for these header entries are described in Table 2. The wavelength for NIR exposure 000 would have the label 'NIR_WL_000'.

$$\lambda = A * (sp - C) - B \tag{1}$$

Table 2: Header entries added at level 1A

Key	**Description**	Type
<channel>_WL_<exp_id>	<channel> TASK <exp_id> wavelength [nm]	String
<channel>_EXP_<exp_id>	<channel> TASK <exp_id> exposure [s]	String

3.2.2 Level 1B

Level 1B processing modifies the data cube by calibrating the raw pixel values into radiance values. This level consist of several steps, each corresponding to a specific calibration procedure. All the calibrations are performed using double-precision values, and the final results are stored as 32-bit floating-point values to a FITS file. The applied calibration steps are described in Table 3.

Table 3: Calibration steps at Level 1B.

Step	**Description**
Float64 conversion	The pixel values are converted to double precision for calculations
CDS-pixel extraction	Extracts CDS pixels from NIR images are extracted and appended to Binary Table extension
Dark background subtraction	Remove dark current based on the master dark frame
Flat field correction	Flat field correction
Bad pixels replacement	Replace bad pixels by neighboring pixel average
Radiometric calibration	Scale the image pixels to radiance by the absolute calibration factor
Float32 conversion	Convert pixel values to float32 for storing

3.2.3 CDS-pixel Extraction

In the original NIR images, the first six rows, the last row, and the four outermost columns on both sides consist of CDS pixels that do not form part of the actual image content [RD3]. These pixels are cropped and stored in a Binary Table extension. Each column in the table corresponds to a single frame and contains a one-dimensional array of 7,984 values, where the pixels are stored in row-major order, with pixels within each row listed from left to right. The column name encodes the channel index and frame number. For example, the column labeled NIR_000 contains the CDS-pixels for frame 000 of the NIR channel. The cropping reduces the NIR image dimension from 648 x 518 pixels to 640 x 512 pixels.

3.2.4 Dark Background Subtraction

Dark background subtraction aims to remove unwanted background signals to enhance the accuracy of subsequent data interpretation. Ideally, an image acquired from a dark frame should consist solely of pixels with an intensity value of zero. However, in most sensors, dark frames include pixels with non-zero values due to dark current forming a background offset. This offset

adds to the light-generated signal during actual illuminated scene imaging. Background noise removal is achieved by subtracting a dark frame from the image. This process entails reducing each pixel value by the corresponding value in the dark frame. Mathematical representation of the dark background subtraction is given in equation 2, where B is the dark background mask.

$$I_{\text{corrected}}(x, y) = I_{\text{raw}}(x, y) - B(x, y) \tag{2}$$

3.2.5 Flat Field Calibration

Images generally exhibit a radial falloff of intensity from the center of the image. This effect is known as vignetting. The calibration process used to correct vignetting in images is called flat field correction. Flat field refers to a uniformly illuminated reference image, which records the sensor's response across all pixels. This reference image is used to adjust the intensity values by dividing each image with it. Mathematical representation of the flat field correction is given in equation 3, where F is the flat field mask.

$$I_{\text{corrected}}(x, y) = \frac{I_{\text{raw}}(x, y)}{F(x, y)} \tag{3}$$

3.2.6 Bad Pixel Removal

Bad pixel refers to a pixel that provides false information due to malfunctioning sensor. Identifying the bad pixels is part of the MIRMIS instrument calibration. The pipeline expects a bad pixel list as a .txt file. An example of the file structure is seen in table 4. The type column indicates the type of the problem: P for pixel, H for horizontal, V for vertical, and R for rectangular cluster. The second and third column indicates the location and the last two rows indicates the size of the affected area. Bap pixels are replaced by averaging the neighboring valid pixels in an iterative loop, such that all clusters are resolved.

Table 4: Bad pixel list example

Type	Col	Row	SizeX	SizeY
P	300	200	1	1
H	-	0	640	1
V	0	-	1	640
R	400	400	3	4

3.2.7 Radiometric Calibration

Radiometric calibration converts raw digital values into radiance units $[\mathrm{W\,m^{-2}\,sr^{-1}\,nm^{-1}}]$ to ensure the accuracy and comparability of the data. To perform the conversion, conversion factors of type ($[\mathrm{DN}/(\mathrm{s} * \mathrm{W\,m^{-2}\,sr^{-1}\,nm^{-1}})]$) are needed for each channel, (order) and wavelength. If an exact match for the wavelength is not found the pipeline estimates the conversion factor by linear interpolating values from the closest wavelengths on each side. The image DN values are then converted using the same formula 4, where I_{exp} is the exposure time is seconds and C is the conversion factor for the wavelength.

$$I_{\text{corrected}}(x, y) = \frac{I_{\text{raw}}(x, y)}{I_{exp} * C} \tag{4}$$

3.2.8 Level 1C

Level 1C converts the radiance values (result of 1B) to I/f reflectance units. The mathematical formula for this conversion is expressed in 5. The pipeline uses average solar spectral irradiance from year 2024, retrieved from NOAA archive [RD10]. The solar spectral irradiance is convolved with Gaussian spectral function (FWHM = 30 nm) and sampled at the instrument exposure wavelengths.

$$\frac{I}{F}(\lambda) = \frac{\pi\, L_\lambda(\lambda)\, r^2}{E_\odot(\lambda, 1\,\mathrm{AU})} \tag{5}$$

In 5, λ denotes the wavelength, $L_\lambda(\lambda)$ is the measured spectral radiance in units of $\mathrm{W\,m^{-2}\,nm^{-1}\,sr^{-1}}$,

and $E_{\odot}(\lambda, 1\,\mathrm{AU})$ is the solar spectral irradiance at 1 AU in units of $\mathrm{W\,m^{-2}\,nm^{-1}}$. The parameter r represents the heliocentric distance of the target expressed in astronomical units, with the factor r^2 accounting for the inverse-square law governing solar illumination. The constant π arises from the definition of reflectance relative to a perfectly Lambertian surface. The resulting quantity $I/F(\lambda)$ is the radiance factor, a dimensionless measure of reflectance that is normalized to the incident solar flux and independent of observation distance.